

Presentation on “Advanced Phishing Detection Using Hybrid A.I. Models”

Submitted To:
Ms. Nidhi Bhatla

Submitted by:
Suhail Kataria(2417163), Arpita Choudhary(2320457),
Rimjhim Sharma(2320487).

Abstract

Phishing attacks are a major cybersecurity threat that exploit both email content and malicious URLs. Traditional detection methods often fail in real-world scenarios due to their reliance on a single type of analysis.

This project proposes a hybrid phishing detection system that combines machine learning with rule-based intelligence to improve detection accuracy and robustness.

➤ System Approach

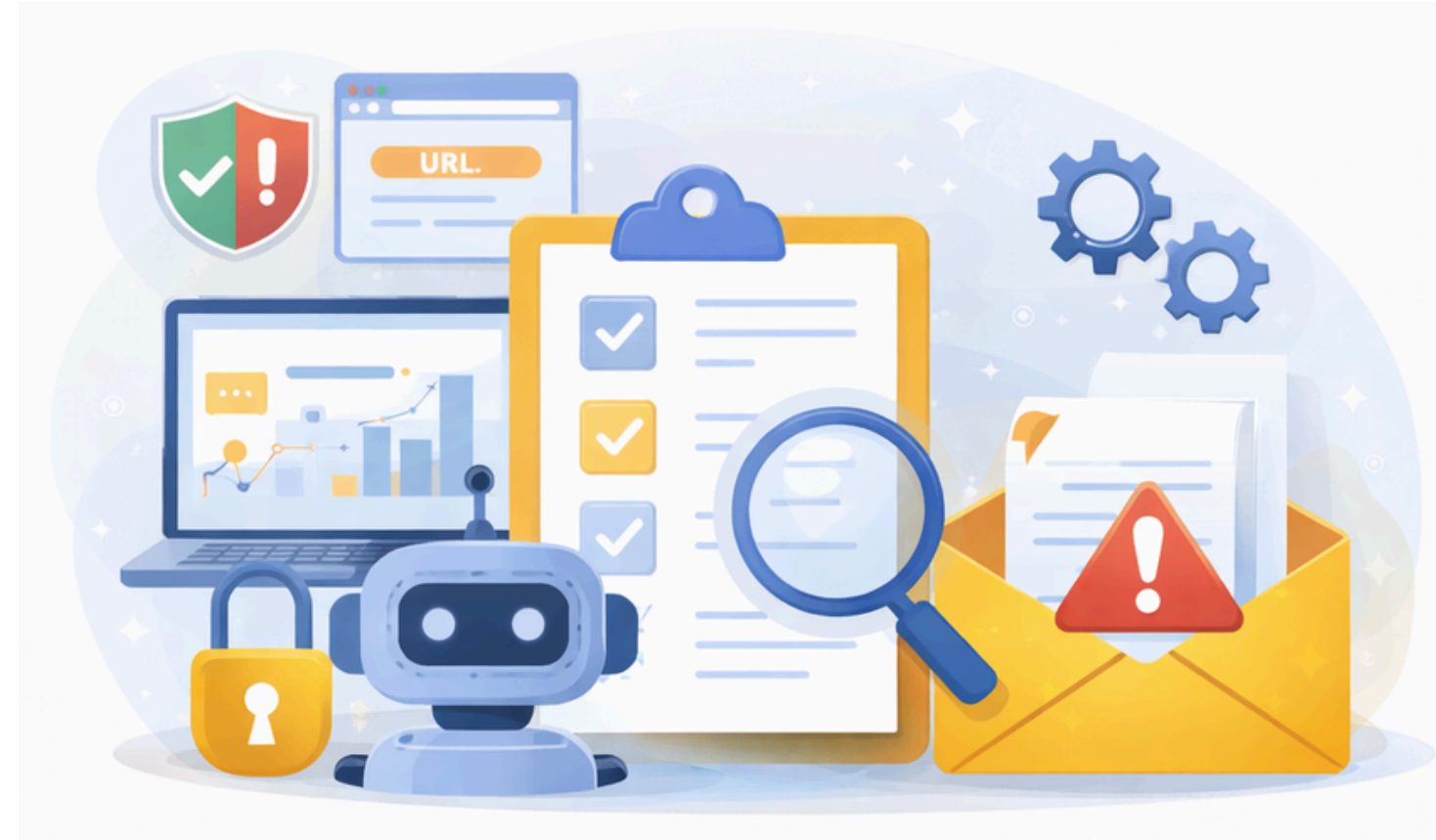
- Transformer-based email analysis (DistilBERT)
- URL structure and pattern analysis
- Joint decision using combined signals

➤ Key Features

- Multi-mode detection:
 - SOC (high recall)
 - Balanced
 - High-confidence
- Explainable results for each prediction
- Batch analysis support

➤ Outcome

- Improved cross-domain detection
- Reduced false positives and false negatives
- Achieved ~90%+ accuracy on evaluation dataset



Introduction

Phishing is one of the most common cyber attacks used to steal sensitive information through fake emails and malicious links.

➤ **Traditional detection systems:**

- rely on either email or URL analysis
- fail in cross-domain scenarios

➤ **Problem:**

- Attackers combine multiple techniques
- Single-model detection is not reliable
- High false positives and false negatives

➤ **Solution**

- Combine email + URL analysis
- Use hybrid detection approach
- Apply rule-assisted decision logic

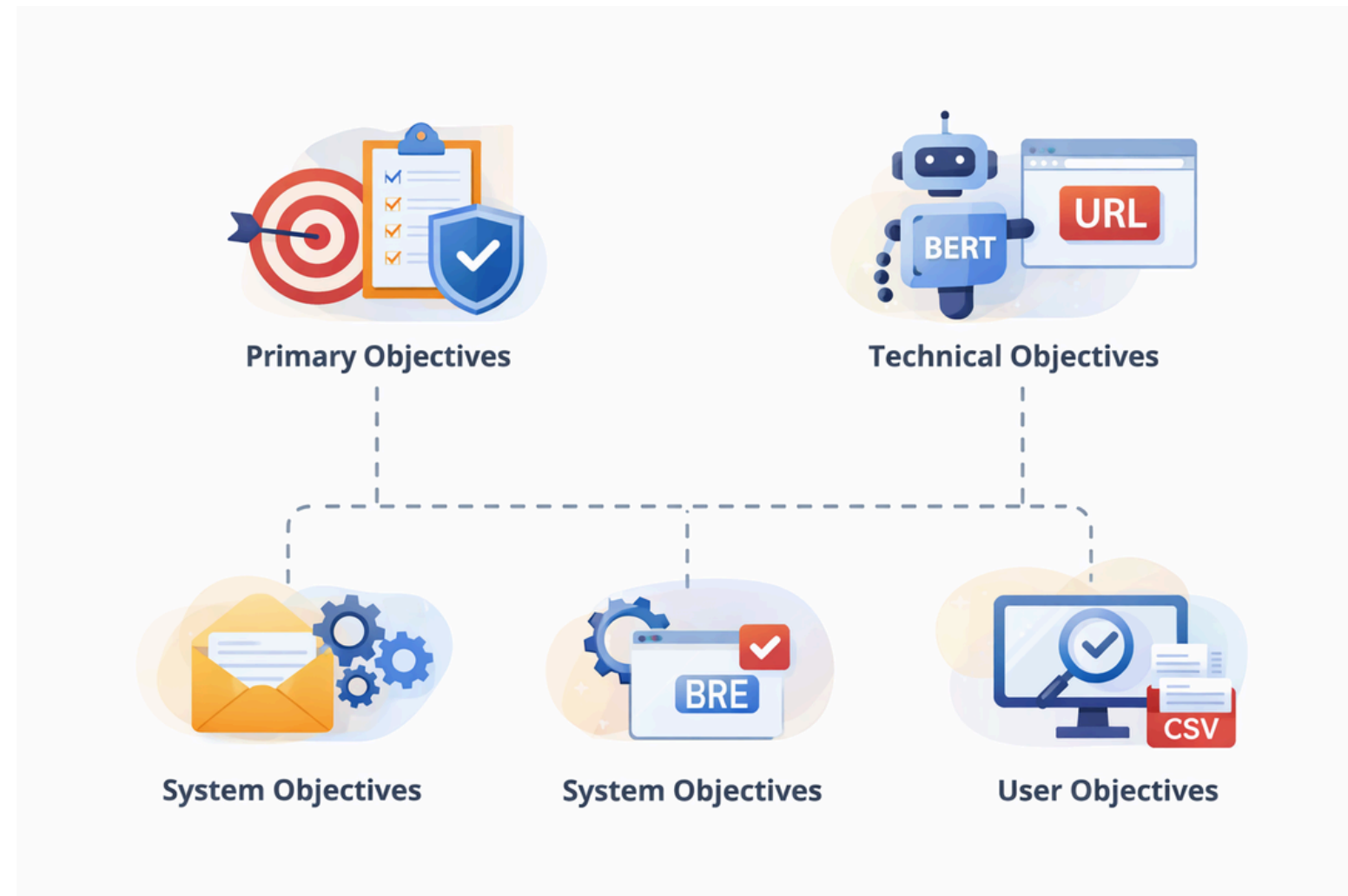
➤ **Motivation**

- Increase detection accuracy
- Improve real-world applicability
- Provide explainable results for users



Objectives of Project

- **Primary Objective:**
 - Develop a robust phishing detection system
 - Improve detection accuracy in real-world scenarios
- **Technical Objectives:**
 - Use transformer-based model for email analysis
 - Analyze URL structure and patterns
- **System Objectives:**
 - Combine email and URL signals
 - Implement rule-assisted decision logic
- **User Objectives:**
 - Provide clear and explainable results
 - Support batch analysis for multiple inputs



Methodology

- **Step 1 — Data Processing**
 - Extract email content (subject + body)
 - Extract URLs from input
 - Clean and prepare data
- **Step 2 — Email Detection**
 - Use DistilBERT transformer model
 - Capture contextual phishing patterns
 - Generate email risk score
- **Step 3 — URL Detection**
 - Analyze domain structure
 - Detect suspicious keywords & patterns
 - Generate URL risk score
- **Step 4 — Joint Detection**
 - Combine email + URL scores
 - Apply rule-based logic:
 - trusted domain validation
 - brand mismatch detection
 - suspicious signal escalation
- **Step 5 — Final Decision:**
 - Apply threshold-based classification
 - Output:
 - Safe / Suspicious / Phishing
 - Explanation + confidence score



Implementation

➤ Tech Stack:

- Frontend: Next.js + Tailwind CSS
- Backend: FastAPI
- ML Model: DistilBERT (Email Detection)

➤ Core Modules:

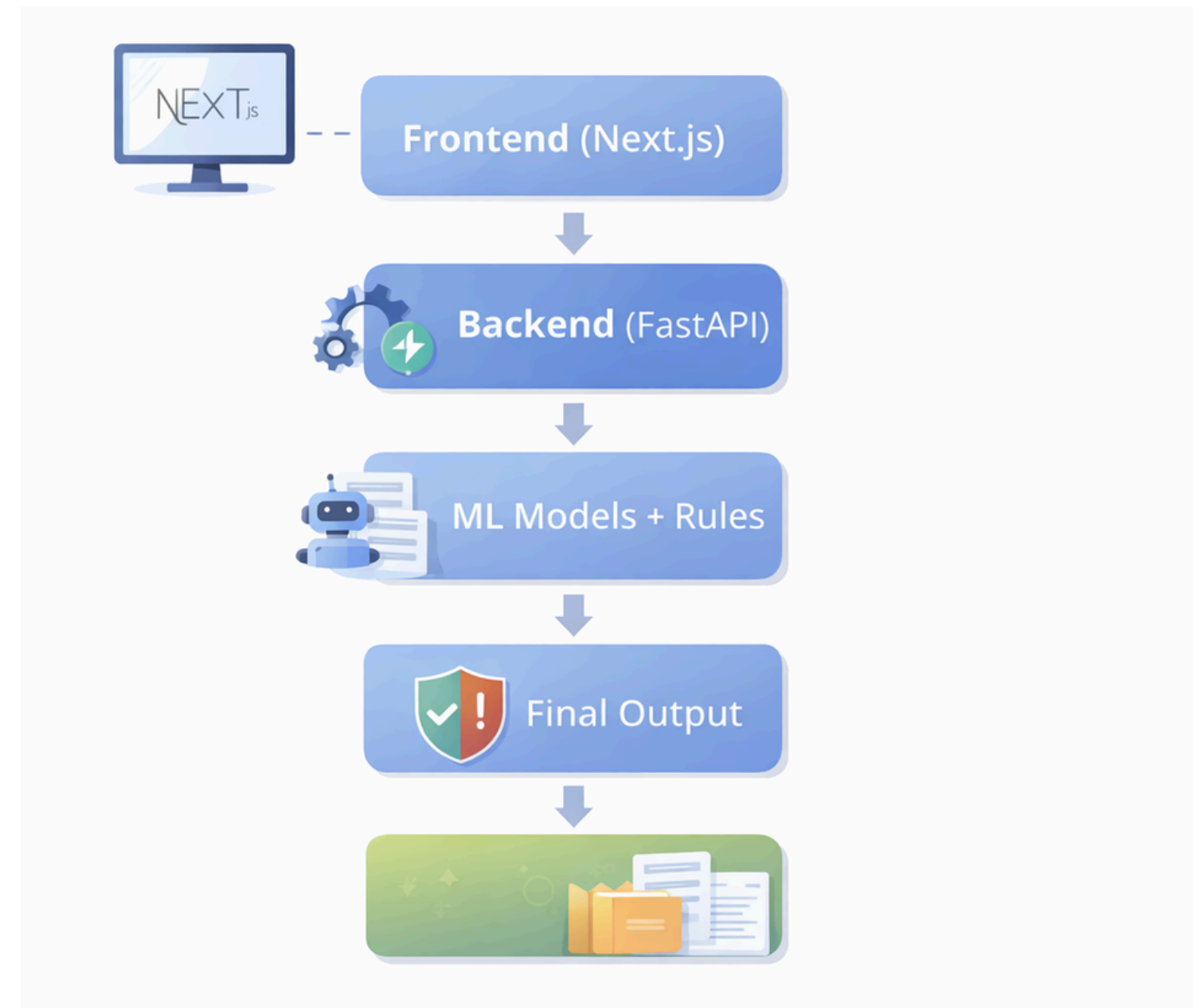
- Email Detection Module
- URL Analysis Engine
- Joint Detection System

➤ Key Features Implemented:

- Transformer-based email classification
- URL structure & pattern analysis
- Rule-assisted joint scoring

➤ Advanced Features:

- Threshold-based detection modes
- Batch CSV analysis
- Explainable output (reason-based results)



Results & Discussion

➤ Model Performance

- Email model → high recall
- URL model → high precision

➤ Final Joint Model

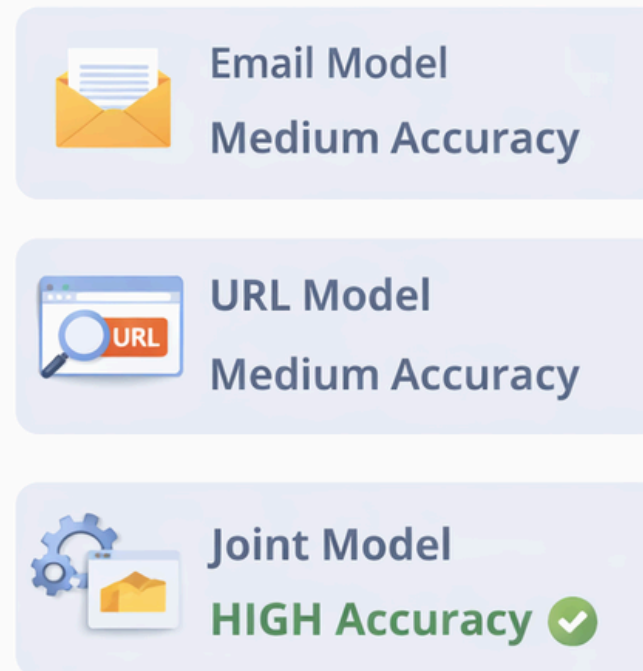
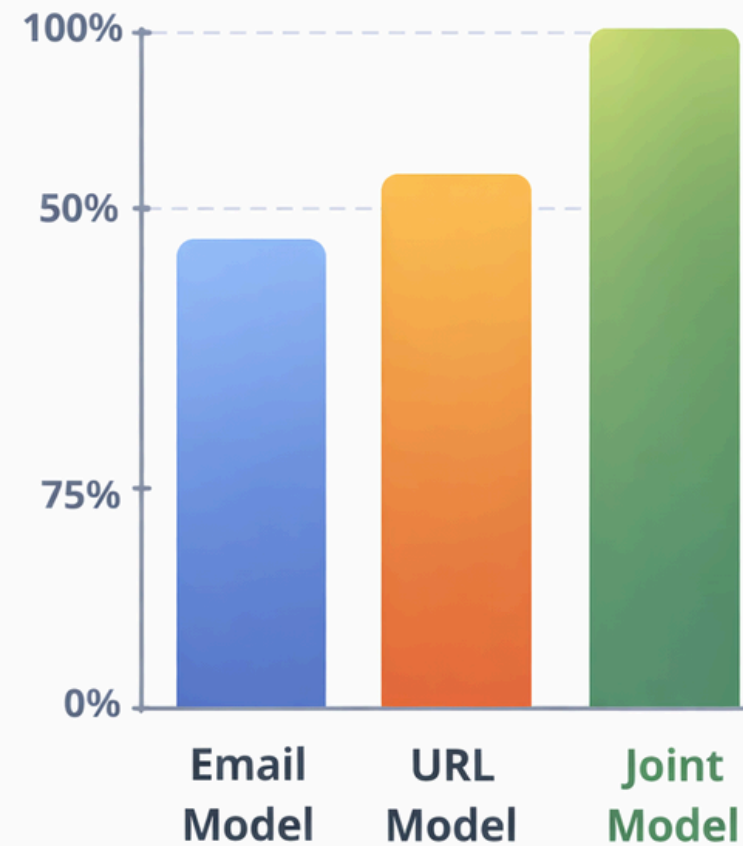
- Accuracy: **~90–94%**
- F1 Score: **~0.94–0.95**

➤ Key Improvements

- Better cross-domain detection
- Reduced false positives
- Reduced false negatives

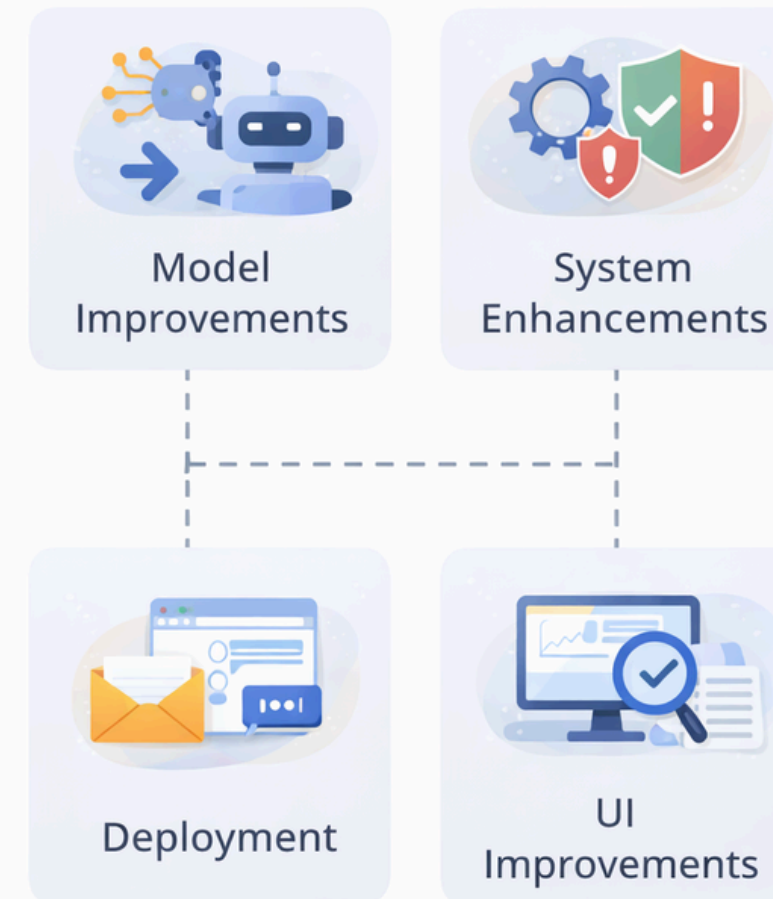
➤ Key Observations

- Hybrid approach outperforms single models
- Threshold tuning improves performance
- Rule-based logic enhances reliability



Future Scope

- **Model Improvements**
 - Use larger real-world datasets
 - Improve zero-day phishing detection
- **System Enhancements**
 - Real-time detection system
 - Integration with threat intelligence
- **Deployment**
 - Browser extension
 - Email client integration
- **UI & User Experience**
 - Advanced dashboard for analysts
 - Improved visualization and reporting



References

➤ Datasets Used:

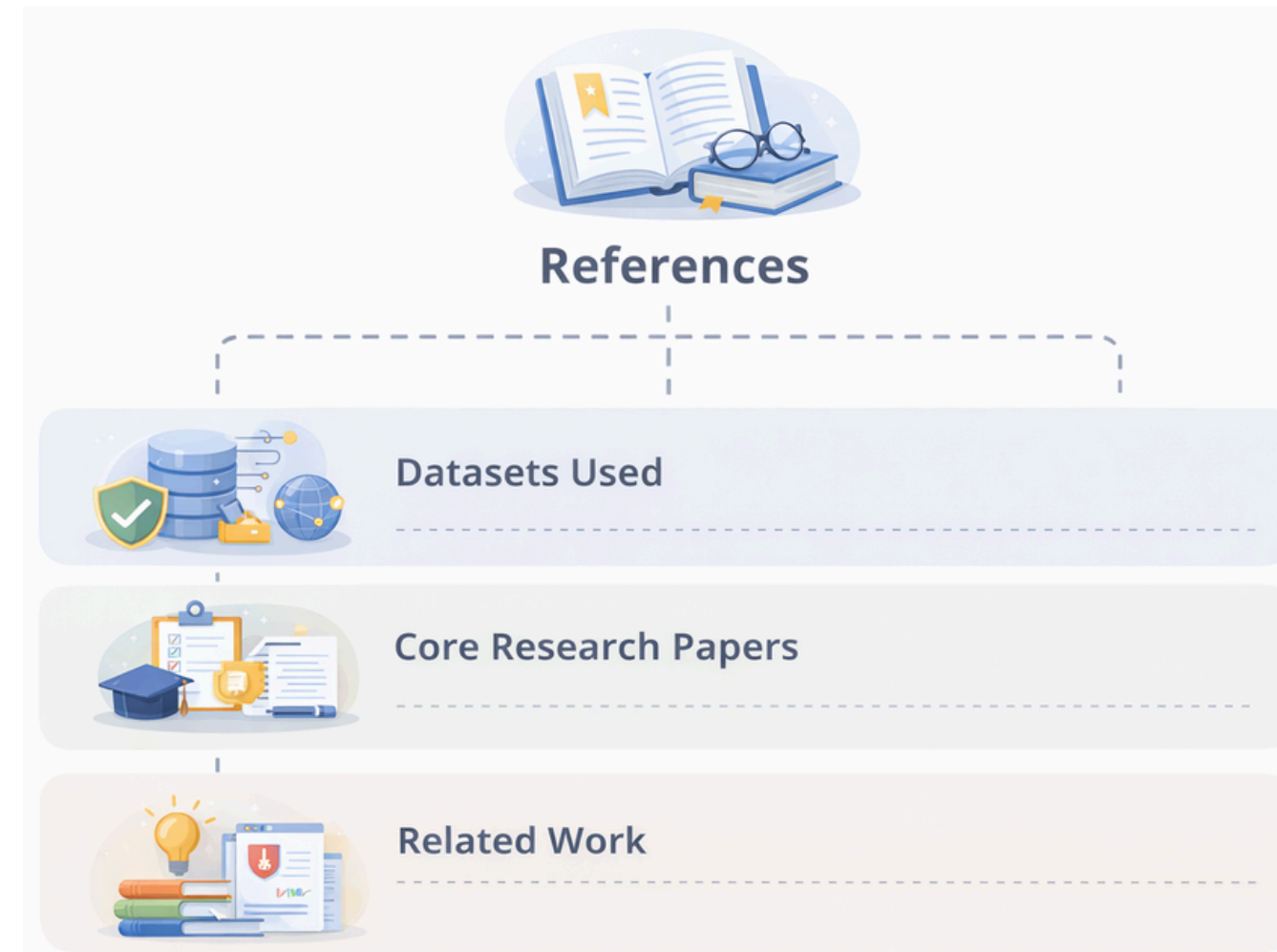
- Enron Email Dataset (Klimt & Yang, 2004)
- SpamAssassin Public Corpus
- Nazario Phishing Corpus
- Benign URLs: Top-1M Dataset
- Phishing URLs: PhishTank Dataset

➤ Core Research Papers:

- Devlin et al., BERT: Language Understanding, 2019
- Sanh et al., DistilBERT Model, 2019
- Metsis et al., Naive Bayes Spam Filtering, 2006
- Fawcett, ROC Analysis, 2006
- Pedregosa et al., Scikit-learn, JMLR

➤ Related Work:

- AI-Based Phishing Detection Techniques (ResearchGate)
- Role of AI in Phishing Detection (Zenodo)
- Hybrid ML Model for Phishing Detection (IEEE)
- Phishing Detection using URL Analysis (IEEE Access)
- Review of AI-Based Phishing Models (IEEE OJ-COMS)



**Thank You
Queries?**